

Probability Measure

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1 Introduction

This lecture dives into probability measures, their mathematical foundations, and key examples.

1.1 Basic Definitions

Definition 1 (Probability Measure) *A probability measure involves:*

- An abstract space Ω (the sample space)
- A corresponding σ -algebra $\mathcal{A}$ containing measurable events

The σ -algebra $\mathcal{A}$ must satisfy:

- Closure under complements: If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$
- Closure under countable unions: If $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$

Definition 2 (Measure) *A measure μ is a function $\mu : \mathcal{A} \rightarrow [0, \infty]$ that is countably additive. For any sequence $\{A_i\}_{i=1}^{\infty}$ of pairwise disjoint sets:*

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Definition 3 (Probability Space) *The triple $(\Omega, \mathcal{A}, P)$ constitutes a probability space when:*

- $P(\Omega) = 1$
- $P : \mathcal{A} \rightarrow [0, 1]$ is a probability measure

2 Examples of Probability Measures

2.1 Discrete Probability Measures

Single Die:

- Sample space: $\Omega = \{1, 2, 3, 4, 5, 6\}$

- σ -algebra: $\mathcal{A} = \mathcal{P}(\Omega)$
- Probability measure: $P(\{i\}) = \frac{1}{6}$ for each $i \in \Omega$

Two Dice:

- Sample space: $\Omega = \{(i, j) : 1 \leq i, j \leq 6\}$
- σ -algebra: $\mathcal{A} = \mathcal{P}(\Omega)$
- Probability measure: $P(\{(i, j)\}) = \frac{1}{36}$ for each $(i, j) \in \Omega$

2.2 Continuous Probability Measures

Normal Distribution

- Sample space: $\Omega = \mathbb{R}$
- σ -algebra: Borel σ -algebra $\mathcal{B}(\mathbb{R})$
- Probability density function:

$$f_X(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Probability measure: For $A \in \mathcal{B}(\mathbb{R})$,

$$P(A) = \int_A f_X(x|\mu, \sigma) dx$$

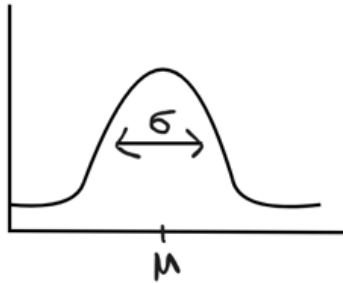


Figure 1: Fig.1 - Normal distribution

3 Types of Probability Measures

3.1 Discrete Measures

Definition 4 (Discrete Measure) A discrete measure on countable space $\Omega = \{x_1, x_2, \dots\}$ with $\mathcal{A} = \mathcal{P}(\Omega)$ satisfies:

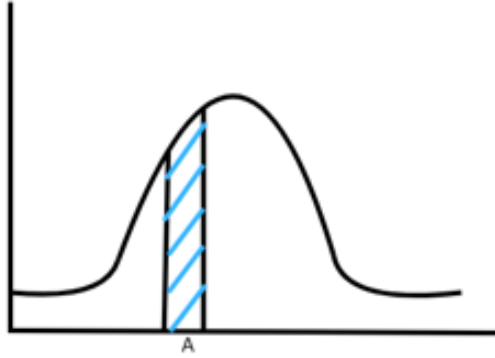


Figure 2: Fig.2 - Normal distribution Integration

- $P(\{x_i\}) = p_i$ with $0 \leq p_i \leq 1$
- $\sum_{i=1}^{\infty} p_i = 1$

For any $A \in \mathcal{A}$:

$$P(A) = \sum_{\{i|x_i \in A\}} p_i$$

Coin Toss: $\Omega = \{H, T\}$ with $P(\{H\}) = p$ and $P(\{T\}) = 1 - p$

3.2 Dirac Measures

Definition 5 (Dirac Measure) For $x \in \mathbb{R}$, the Dirac measure δ_x on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is:

$$\delta_x(A) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{otherwise} \end{cases}$$

Proposition 6 Any discrete measure on $\mathbb{R}$ can be expressed as a sum of Dirac measures:

$$P = \sum_{i=1}^n p_i \delta_{x_i}$$

The die measure from Example 2.1 can be written as:

$$P = \frac{1}{6}(\delta_1 + \delta_2 + \cdots + \delta_6)$$

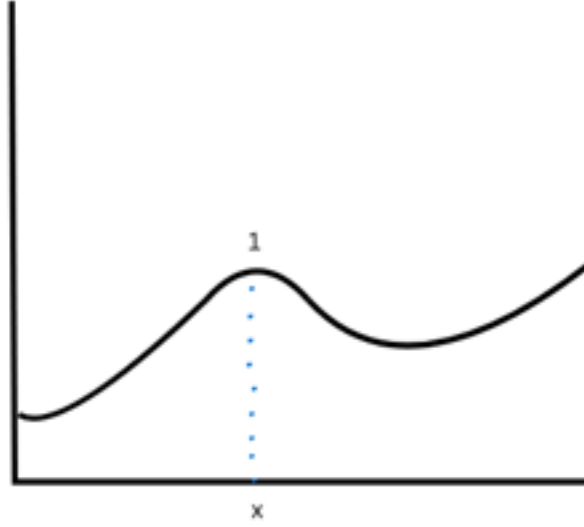


Figure 3: Fig.3 - Dirac Measure Figure

4 Probability Measures with Density Functions

Definition 7 (Measure with Density) In the context of the measure space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, where λ is the Lebesgue measure, a measurable function $f : [0, \infty]$ satisfying:

$$\int_{\mathbb{R}} d\lambda = 1 \quad \text{which is} \quad \int_A f(x)dx = 1$$

defines a probability measure γ on $\mathbb{R}^n$ via:

$$\gamma(A) = \int_A f(x)dx \quad \text{for all } A \in \mathcal{A}$$

γ is the probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ with density f . We denote this relationship as $\gamma = f\lambda$.

Remark 8 Not all probability measures on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ admit density functions. For example:

- The Dirac measure δ_x cannot be expressed as $\delta_x = f\lambda$ for any f
- Discrete measures like $\sum_{i=1}^n p_i \delta_{x_i}$ lack density functions

5 Absolute Continuity of Measures

Definition 9 (Absolute Continuity) A probability measure γ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ is absolutely continuous with respect to μ ($\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n)$) if for every μ -null set is also a γ -null set : $\forall B \in \mathcal{B}(\mathbb{R}^n) :$ if $\mu(B) = 0$ then $\gamma(B) = 0$

$$\mu(A) = 0 \implies \gamma(A) = 0 \quad \forall A \in \mathcal{B}(\mathbb{R}^n)$$

Notation; $\gamma \ll \mu$

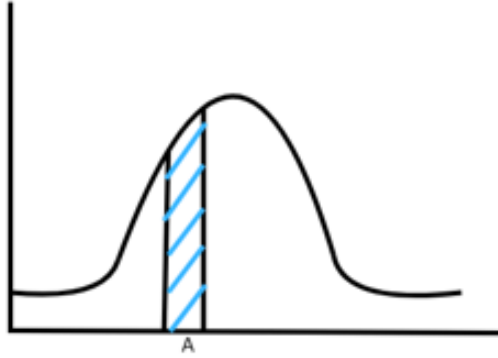


Figure 4: Fig.4 - Integral of continuous distribution

6 Radon-Nikodym Theorem

Theorem 10 (Radon-Nikodym) For σ -finite measures μ and γ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, the following are equivalent:

1. γ has a density with respect to μ (i.e., $\exists f$ measurable with $\gamma(A) = \int_A f d\mu$)
2. $\gamma \ll \mu$: γ is absolutely continuous with respect to μ

If $\gamma \ll \mu$, Then $\exists \phi$ such that, $\gamma(A) = \int_A \phi d\mu$ Then, ϕ exists and is unique.

6.1 Proof:

Step 1: Define the Set of Dominating Functions

Define the set G of all measurable functions $g : X \rightarrow [0, \infty)$ satisfying

$$\int_A g d\mu \leq \gamma(A) \quad \forall A \in \mathcal{B}.$$

- G is non-empty because the zero function $g \equiv 0$ belongs to G .
- If $g_1, g_2 \in G$, then $\max(g_1, g_2) \in G$ since for any A , we can split A into $A_1 = \{g_1 \geq g_2\}$ and $A_2 = \{g_1 < g_2\}$, leading to

$$\int_A \max(g_1, g_2) d\mu \leq \gamma(A).$$

Step 2: Define the Supremum and Construct a Maximizing Sequence

Define

$$X = \sup_{g \in G} \int_X g d\mu.$$

Since γ is σ -finite, we have $X < \infty$. Choose a sequence $\{g_n\} \subset G$ such that

$$\lim_{n \rightarrow \infty} \int_X g_n d\mu = X.$$

Define the increasing sequence

$$f_n = \max(g_1, \dots, g_n),$$

which satisfies $f_n \in G$ and $\lim_{n \rightarrow \infty} \int_X f_n d\mu = X$ by construction.

Step 3: Construct the Density Function

Define

$$f = \sup_n g_n.$$

By the monotone convergence theorem,

$$\int_A f d\mu = \sup_n \int_A g_n d\mu \leq \gamma(A) \quad \forall A \in \mathcal{B}.$$

However, since $f_n \in G$ and $\int_X f_n d\mu$ approaches the supremum X , we conclude

$$\int_A f d\mu = \gamma(A),$$

ensuring f is the required Radon–Nikodym derivative.

Step 4: Uniqueness

Suppose there exist two such functions f_1 and f_2 . Then

$$\int_A (f_1 - f_2) d\mu = 0 \quad \forall A \in \mathcal{B}.$$

Let $A = \{f_1 > f_2\}$. Then $\mu(A) = 0$ since $\int_A f_1 d\mu = \int_A f_2 d\mu$. Similarly, $\mu(\{f_2 > f_1\}) = 0$, implying $f_1 = f_2$.

Thus, the function f is unique up to sets of measure zero, completing the proof.

7 Bertrand paradox

Probability that a random chord exceeds $\sqrt{3}$ Consider the unit circle and the length of a random chord. The probability that the chord length $L > \sqrt{3}$ depends on how we define "random":

1. **Uniform endpoint method (Probability = 1/3):** Fix one endpoint and choose the other uniformly. The chord length exceeds $\sqrt{3}$ when the angle $\theta \in (2\pi/3, 4\pi/3)$.
2. **Uniform radius method (Probability = 1/2):** Choose a random radius and a point on it uniformly for the chord. The condition holds when the distance $d < 1/2$ from center.
3. **Uniform midpoint method (Probability = 1/4):** Choose the midpoint uniformly in the disk. The condition holds when the midpoint lies in a circle of radius $1/2$.

这反映了概率论的一个基本原则：概率空间 $(\Omega, \mathcal{F}, P)$ 必须明确定义。

不同的随机实验对应不同的 σ -代数和概率测度。

在应用中，“无差别原则” (principle of indifference) 必须小心使用，否则会导致矛盾。

We calculate each case:

1. **Endpoint method:**

$$P(L > \sqrt{3}) = P(\cos \theta < -1/2) = \frac{4\pi/3 - 2\pi/3}{2\pi} = \frac{1}{3}$$

2. **Radius method:** The length condition becomes $2\sqrt{1-d^2} > \sqrt{3} \Rightarrow d < 1/2$. Thus:

$$P = \frac{1/2}{1} = \frac{1}{2}$$

3. **Midpoint method:** The area where $L > \sqrt{3}$ is a circle of radius $1/2$:

$$P = \frac{\pi(1/2)^2}{\pi(1)^2} = \frac{1}{4}$$

Discussion

This example illustrates the **Bertrand paradox**, showing how probability results depend critically on the selected defined sample space. The different probabilities $(1/2, 1/3, 1/4)$ correspond to different natural σ -algebras and measures on the space of chords.

8 Applications in Artificial Intelligence

Probability measures play a crucial role in AI, particularly in probabilistic modeling, Bayesian inference, and deep learning. Many AI systems rely on probability distributions to model uncertainty and make informed decisions.

Fact: Gaussian Processes in Machine Learning: Gaussian Processes (GPs) leverage probability measures to define distributions over functions. This enables AI models to perform regression and classification tasks with built-in uncertainty. GPs are widely used in Bayesian optimization, reinforcement learning, and time-series forecasting.